

AI-Based Multimodal Retinal & Clinical Analysis System

Multimodal Retinal & Clinical Assessment Report — v1.0.0 (Research Edition)

Patient / Sample ID:	TEST_API_PATIENT	Report ID:	REP-TEST_API_PAT-20260829111927
Assessment Date:	2026-08-29 11:19:27	Input Modalities:	OCTA

1. Patient Clinical Profile & Biomarkers

Age / Group	O_CD	Gender	Male (1)
Education (Years)	16.0	BMI (kg/m²)	26.5
Hypertension (HTN)	Positive (1)	Diabetes (DM2)	Negative (0)
Smoking History	History (1)	Alcohol History	History (1)

2. Retinal Image Technical Quality (Phase 3)

Modality	Status	Quality Score	Technical Decision
OCTA	Available	85.0 / 100	ACCEPT
OCTB	Not available	—	—
FUNDUS	Not available	—	—

3. Multi-Task Disease Predictions & Uncertainty Analysis

Assessment Parameter	Stroke Target	Alzheimer's Disease Target
Model Prediction Class	Class 1 (Positive)	Class 0 (Negative)
Predicted Probability	0.5232	0.4507
Research Risk Category	MODERATE RISK	MODERATE RISK
Model Confidence (Phase 9)	99.54% (HIGH CONFIDENCE)	99.38% (HIGH CONFIDENCE)
Predictive Uncertainty (Variance)	$\sigma^2 = 0.0010$ (LOW)	$\sigma^2 = 0.0010$ (LOW)
Predictive Shannon Entropy	$H(p) = 0.6902$ nats	$H(p) = 0.6875$ nats

4. Retinal Visual Explainability (Swin Grad-CAM)

Grad-CAM heatmaps highlight salient retinal regions contributing to model logits. Each panel displays [Original Retinal Scan | Grad-CAM Heatmap | Alpha-Blended Overlay].

Retinal Grad-CAM visualizations not available for this session.

5. Clinical Feature Attributions (Game-Theoretic SHAP)

Shapley values (SHAP) quantify marginal additive feature contributions to model log-odds.

Feature Name	Patient Value	Stroke SHAP (Impact)	Alzheimer's SHAP (Impact)
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HTN	1	-0.0165 (DECREASES_RISK)	+0.0182 (INCREASES_RISK)
BMI	26.5	-0.0164 (DECREASES_RISK)	-0.0043 (DECREASES_RISK)
Smoking_ever	1	+0.0110 (INCREASES_RISK)	+0.0035 (INCREASES_RISK)
Gender	1	-0.0110 (DECREASES_RISK)	+0.0321 (INCREASES_RISK)
DM2	0	+0.0109 (INCREASES_RISK)	+0.0265 (INCREASES_RISK)
Obese	0.0	-0.0101 (DECREASES_RISK)	-0.0019 (DECREASES_RISK)
Education	16.0	+0.0089 (INCREASES_RISK)	+0.0158 (INCREASES_RISK)
EtOH_ever	1	+0.0034 (INCREASES_RISK)	-0.0145 (DECREASES_RISK)

6. Unified Multimodal Synthesis & Findings

Multimodal AI assessment was performed for patient 'TEST_API_PATIENT' utilizing available retinal modalities (OCTA) and patient tabular clinical health variables. For the Stroke target, the model estimated a predicted probability of 0.5232 (MODERATE RISK), with an estimated model confidence of 99.5% and low predictive uncertainty (variance=0.0010). For the Alzheimer's Disease target, the model estimated a predicted probability of 0.4507 (MODERATE RISK), with an estimated model confidence of 99.4% and low predictive uncertainty (variance=0.0010). Model explainability heatmaps (Grad-CAM) and game-theoretic clinical feature contributions (SHAP) are integrated to highlight computational saliency patterns.

7. Research Limitations & Methodological Constraints

- 1. Experimental Model Scope: Algorithms are trained on curated academic datasets and require prospective clinical validation.
- 2. Biological Complexity: Retinal microvascular and neurodegenerative patterns reflect multi-factorial systemic processes.
- 3. Explainability Attribution: Grad-CAM heatmaps and SHAP values indicate model parameter attributions and do not establish medical causality.
- 4. Modality Missingness: Missing imaging modalities degrade fusion representational capacity relative to complete tripartite scans.

MANDATORY RESEARCH DISCLAIMER:

IMPORTANT RESEARCH NOTICE: This report is generated by an experimental multimodal artificial intelligence research prototype for investigational and decision-support purposes only. The predictions, probability estimates, uncertainty metrics, and explainability visualizations do not constitute a clinically verified medical diagnosis and must never substitute for professional evaluation, diagnosis, or treatment by a board-certified healthcare provider.